Tutorial Questions (Set-IV)

Q1 Let (X, d) be a metric space and K be a subset of X.

(i) If K is compact, then show that K is sequentially compact.

(ii) If X is compact (or sequentially compact) and K is closed, then show that K is compact (resp. sequentially compact).

(iii) If K is compact and F is a closed subset of X, then show that $K \cap F$ is compact.

(iv) Let $f\colon (X, d) \to (Y, \rho)$ be a continuous function. If K is compact (or sequentially compact) subset of X, then show that the image $f(K)$ is compact (resp. sequentially compact) subset of Y.

(v) If $f\colon (X, d) \to (Y, \rho)$ is a bijective continuous function and X is compact. then show that f is a homeomorphism.

Q2 Let (X_1, d_1) and (X_2, d_2) be metric spaces. Then d_E given by

$$d_E((x_1, x_2), (y_1, y_2)) = \sqrt{(d_1(x_1, y_1))^2 + (d_2(x_2, y_2))^2}, \quad (x_1, x_2), (y_1, y_2) \in X_1 \times X_2$$

is a metric on the cartesian product $X_1 \times X_2$.

(i) If $A \subseteq X_1$ and $B \subseteq X_2$ are sequentially compact subsets, then show that $A \times B$ is a sequentially compact subset of $(X_1 \times X_2, d_E)$.

(ii) Show that an n-cube $[a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n]$ for $a_i < b_i \ \ (1 \leq i \leq n)$, is a sequentially compact (hence, a compact) subset of Euclidean space $\mathbb{R}^n$.

Q3 Let (X, d) be a compact metric space and $f\colon X \to \mathbb{R}$ be a continuous function.

(i) Show that f is bounded and f attains its bounds.

(ii) Show that f is uniformly continuous.

(iii) If X is also connected, then show that $f(X)$ is a closed and bounded interval.

(iv) Show that every continuous map $\varphi\colon [0, 1] \to [0, 1]$ has a fixed point.

Q4 Let (X, d) be a metric space and $K \subseteq X$.

(i) If K is compact, then show that K is totally bounded.

(ii) Show that a totally bounded subset is bounded.

(iii) Give an example of bounded subset that is not totally bounded.

(iv) Show that every bounded subset of Euclidean space $\mathbb{R}^n$ is totally bounded.

(v) (**Optional**) If X is compact, then show that X has a countable dense subset.

Q5 Let (X, d) be a metric space, $K \subseteq X$ and $\{G_\alpha : \alpha \in I\}$ be an open cover of K. If K is compact, then show that the given open cover has a Lebesgue number.

Q6 Suppose $f\colon (a, b) \to \mathbb{R}$ and $g\colon (a, b) \to \mathbb{R}$ are real-valued functions on an open interval (a, b) such that

$$\lim_{t \to x} f(t) = A \quad \text{and} \quad \lim_{t \to x} g(t) = B \quad \text{for some } x \in (a, b).$$

Show that

(i) $\lim_{t \to x}(f \pm g)(t) = A \pm B$.

(ii) $\lim_{t \to x}(fg)(t) = AB$.

(iii) $\lim_{t \to x} \left(\frac{f}{g}\right)(t) = \frac{A}{B}$, provided $g(t) \neq 0$, $\forall t \in (a, b)$ and $B \neq 0$.

Hence, deduce that if f and g are continuous at x, then so are functions $f \pm g, fg$ and $\frac{f}{g}$, provided $g(t) \neq 0$ for all $t \in (a, b)$ in the last case.

Q7 (i) Let E be a countable dense subset of an open interval (a, b). Construct a monotonic function $f\colon (a, b) \to \mathbb{R}$ such that the set of discontinuities of f is precisely E.

(ii) A rational number $x \in \mathbb{Q}$ can be expressed in the standard form $x = \frac{p}{q}$, where $p \in \mathbb{Z}, q \in \mathbb{N}$ and $\gcd(p, q) = 1$. If $x = 0$, we take $q = 1$. Define a function $g\colon \mathbb{R} \to \mathbb{R}$ given by

$$g(x) = \begin{cases} \frac{1}{q}, & \text{if } x = \frac{p}{q} \in \mathbb{Q};\ x \text{ in standard form,} \\ 0, & \text{otherwise.} \end{cases}$$

Show that g is continuous at every $x \in \mathbb{R} \setminus \mathbb{Q}$. Further, show that

$$g(x+) = g(x-) = 0 \quad \text{for} \quad x = \frac{p}{q} \in \mathbb{Q}.$$

(The function g is called *Thomae function* or *modified Dirichlet function.*)

(iii) Is it possible to construct a function $h\colon \mathbb{R} \to \mathbb{R}$ having discontinuities precisely at all irrational numbers? Explain.